

Extracting Multiword Expressions and Evaluating Google Translate Using a Parallel English–Arabic Corpus

Doha Hashish

matriculation number: 6936778

1. Introduction

1.1. What MWEs Are

“Multi-word expressions “MWEs” are linguistic objects formed by two or more words that behave like a ‘unit’ by displaying formal and/or functional idiosyncratic properties with respect to free word combinations. They include an extremely varied set of items (from idioms to collocations, from formulae to sayings) which have been the privileged subject matter of fields such as phraseology, lexicology, lexicography, and computational linguistics.” Masini (2019), Oxford Research Encyclopedia of Linguistics.

1.2. Compound Vs MWEs

A compound is a word formed by combining two or more bases, roots, or stems into a single lexical unit.

A multi-word expression (MWE) is a lexical item made up of more than one word that functions as a single semantic or pragmatic unit.

Compounds and multi-word expressions overlap because both act as multi-word lexical units, and since linguists disagree on how to separate them from each other or from ordinary phrases, there is no single accepted definition for either category in English.

1.3. MWEs vs Collocations

Property	MWEs	Collocations
Meaning	Often non-compositional (cannot be predicted from parts)	Compositional (meaning predictable from parts)
Fixedness	High — fixed order, not reversible, not interruptible	Low–medium — preferred combinations but flexible
Lexicalization	Stored as single lexical units	Not lexicalized; just frequent co-occurrence
Variability	Very restricted	More flexible (can vary with modifiers, tense, etc.)
Syntactic freedom	Often syntactically frozen	Fully syntactic and productive

Examples	• kick the bucket → ‘die’ • by and large → ‘generally; on the whole’ • salt and pepper → ‘a fixed pair; also a color term meaning mixed black and white’ • look up to → ‘admire; respect’ • touch and go → ‘uncertain; risky; could go either way’ • lady-in-waiting → ‘female attendant to a queen or princess’	<i>strong tea, heavy rain, make a decision, utter nonsense, take a picture</i>
Test	Changing a word breaks the meaning (<i>kick the pail</i> ≠ “die”)	Changing a word still works (<i>powerful tea, big rain</i>)
Source of fixedness	Idiomacity, cultural convention, lexicalization	Statistical frequency, usage patterns

1.4. Why MWEs are Hard to Translate

- MWEs are often semantically opaque (non-compositional); The meaning cannot be predicted from the individual words.
- MWEs are culturally bound; Many MWEs rely on cultural knowledge, metaphors, or historical references that do not exist in the target language.
- MWEs have fixed lexical and syntactic behavior
- MWEs often have no direct equivalent in the target language; Languages carve up meaning differently. English may lexicalize a concept that Arabic expresses differently
- MWEs mix literal and idiomatic meanings; Some MWEs have both: salt and pepper (food literal) vs. (hair color idiomatic) red herring (literal fish) vs. (idiomatic “misleading clue”)

1.5. What the Project Aims to Evaluate

This project evaluates how accurately Google Translate renders English multi-word expressions (MWEs) into Arabic by comparing its output to official UN human translations. The goal is to identify where Google

succeeds, where it fails, and what types of errors appear most frequently. By applying the MQM error framework to each MWE, the project examines Google's handling of terminology, idiomatic meaning, context-dependent expressions, and stylistic naturalness. The evaluation ultimately shows how well machine translation deals with MWEs in a real, domain-specific setting and where human translators still outperform automated systems.

2. Extracting MWEs from English

MWEs were extracted from the English corpus using AntConc's N-gram tool to identify fixed multiword sequences. The 2-, 3-, 4-, 5-, and 6-gram outputs were saved in a combined Excel file ("2, 3, 4, 5, 6 gram results"). Only the first 100 items from each list were kept, since frequency drops sharply after that point (for example, the last 4-gram in the top-100 appears only 15 times).

A refined MWE list was then produced ("2, 3, 4, 5, 6 gram results refined.xlsx"). All N-grams were merged, alphabetically sorted, and deduplicated. A manual cleaning step removed noise such as function-word bundles (e.g., *of the, in the*) and frequent but non-idiomatic combinations (e.g., *the cost, october expenditure november october*).

Several extraction issues were observed. AntConc ignores numbers inside MWEs, so expressions like *April expenditure* actually appear in the corpus as *April 1992 expenditure*. Because the numerical element is dropped, the extracted string does not exist in the parallel corpus and therefore produces zero matches. AntConc also does not preserve line breaks when generating N-grams, which leads to merged sequences that never occur as continuous phrases. For example, it produced *assembly security council forty seventh session* even though the corpus contains these elements on separate lines:

GENERAL ASSEMBLY SECURITY COUNCIL

Forty seventh session Forty eighth year

Since these words come from different lines, the merged N-gram does not exist in the parallel corpus. Finally, punctuation is ignored, which affects expressions such as *Spare parts, repairs and maintenance*, extracted incorrectly as *parts repairs and maintenance*.

3. Evaluation Method

The evaluation of Google Translate's handling of multiword expressions (MWEs) uses the [MQM Core Typology](#), which offers a systematic and fine-grained way to identify translation errors. Each English MWE extracted from the corpus is aligned with its UN human Arabic translation and Google's Arabic output and then assessed according to MQM categories relevant to

MWE behavior. These include Accuracy (mistranslation, undertranslation, addition, omission, or cases where the expression should not be translated), Terminology (inconsistent term choice), Linguistic Conventions (grammatical issues and violations of Arabic writing norms), and Style (register problems, awkward phrasing, and unidiomatic expressions). Together, these categories capture typical MT difficulties such as loss of idiomatic meaning, incorrect lexical choices, and stylistic unnaturalness.

MQM is used because it is a transparent, widely recognized, and replicable error typology. It allows consistent evaluation across different error types and is well suited for analyzing the complex, idiomatic, and context-dependent nature of MWEs in machine translation output.

The following table lists the MQM Core Typology categories that are relevant to MWE evaluation, extracted directly from the official MQM framework and limited to the error types applicable to this project.

MQM Category	Subcategory	Meaning of Error (Definition)
Terminology	Inconsistent use of terminology	Same term translated differently across occurrences.
Terminology	Wrong term	Incorrect term chosen for the meaning.
Accuracy	Mistranslation	Target meaning does not match source meaning.
Accuracy	Undertranslation	Only part of the meaning is conveyed.
Accuracy	Addition	Extra meaning added that was not in the source.
Accuracy	Omission	Meaning partially or fully missing.
Accuracy	Do not translate	Translates something that should remain unchanged.
Linguistic conventions	Grammar	Errors in grammatical structure.
Linguistic conventions	Textual conventions	Violations of target-language writing norms.
Style	Language register	Wrong level of formality.
Style	Awkward style	Grammatically correct but stylistically clumsy.
Style	Unidiomatic style	Not natural or idiomatic in the target language.

4. Results

4.1 Human vs. Google Translation Examples

The comparison between UN human translations and Google Translate outputs shows that Google correctly renders most MWEs in the dataset. Out of all analyzed items, 29 were classified as “Correct term,” indicating that Google often selects an acceptable Arabic equivalent. However, the remaining cases reveal clear weaknesses in specific areas such as official terminology, legal and financial expressions, idiomatic usage, and stylistic conventions.

Examples include mistranslations of specialized terms (e.g., jurisdiction, provision), incorrect renderings of official entity names, and literal translations of idiomatic expressions. These illustrate that while Google performs well on general MWEs, it struggles when contextual or institutional knowledge is required.

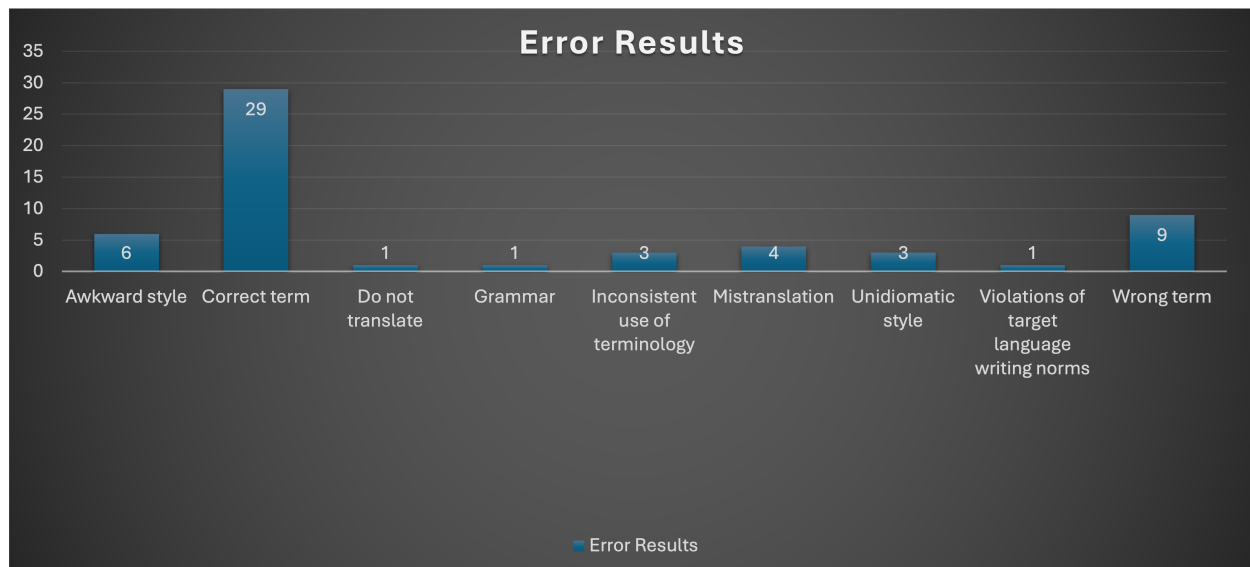
4.2 Error Types

The MQM analysis identified several error categories in Google’s output: wrong term, inconsistent terminology, mistranslation, unidiomatic style, awkward style, grammar errors, violations of Arabic writing norms, and incorrect translation of metadata that should remain in English. These errors reflect both linguistic and contextual limitations in Google’s handling of UN-style texts.

4.3 Patterns Observed

Several patterns emerged across the dataset. Google struggled with official UN terminology and context-dependent expressions, often producing literal or stylistically unnatural Arabic. In contrast, Google showed higher consistency than the UN in MWEs that have multiple acceptable Arabic equivalents. Style and register issues were frequent, and context sensitivity was limited, especially in legal, financial, and diplomatic terminology.

4.4 Statistics



The chart shows the frequency of each MQM error type identified in Google's translations of the extracted MWEs. The largest category is Correct term (29 instances), indicating that Google successfully renders most MWEs. Errors are concentrated in specific areas: Wrong term (9) and Mistranslation (4) occur mainly in legal, financial, and institutional contexts. Awkward style (6) reflects issues with naturalness and phrasing. Unidiomatic style (3) and Inconsistent terminology (3) appear with equal frequency, showing challenges in idiomatic usage and terminological consistency. Less common issues include Grammar (1), Violations of writing norms (1), and Do not translate (1). Overall, the distribution demonstrates that while Google performs well on many MWEs, its errors cluster around context-dependent and domain-specific expressions.

5. Discussion

The results show that Google handles most MWEs correctly, but its mistakes reveal exactly where MT still struggles. The system performs well when the expression has a common, widely used Arabic equivalent. Once the meaning depends on legal, financial, or UN-specific context, Google starts choosing the wrong terms or producing unnatural phrasing. These errors happen because MT relies on statistical patterns, not on institutional knowledge or discourse conventions. Human translators, on the other hand, know how the UN uses terminology, understand the legal and administrative context behind each expression, and choose equivalents that fit the register and the document type. The contrast between the high number of correct translations and the

concentrated errors shows that MT is strong on surface-level language but still weak on deeper contextual interpretation.

6. Conclusion

This project shows that Google Translate can correctly render many MWEs, but its errors consistently appear in areas where context and domain knowledge matter. Human translators outperform MT in legal, financial, and institutional terminology because they understand how meaning shifts across contexts and how the UN standardizes its language. Overall, the findings highlight that MT is reliable for general expressions but still needs improvement in handling specialized vocabulary, context-dependent meanings, and natural Arabic style.

Technical Appendix:

Data & Tools

The English corpus and its official human Arabic translation were taken from the [UN Parallel Corpus](#), which provides line-aligned multilingual UN documents. [AntConc](#) was used to extract the N-grams that form the initial list of candidate MWEs. [AntPConc](#) was then used to build a parallel corpus consisting of the first 5,000 line breaks of the UN English text, its UN human Arabic translation, and the Google-translated Arabic version. This three-row parallel setup allowed direct comparison between human and machine translation for each extracted MWE.

Plain-text bitexts

We also make available plain-text bitexts that span all documents for a specific language pair and can be used more readily with SMT training pipelines. Inside a language-pair specific archive consists of a plain-text file for each language and one file with ids.

	French	Spanish	Russian	Chinese	Arabic
English	UNv1.0.en-fr.tar.gz.00	UNv1.0.en-es.tar.gz.00	UNv1.0.en-ru.tar.gz.00	UNv1.0.en-zh.tar.gz.00	UNv1.0.ar-en.tar.gz.00
	UNv1.0.en-fr.tar.gz.01	UNv1.0.en-es.tar.gz.01	UNv1.0.en-ru.tar.gz.01	UNv1.0.en-zh.tar.gz.01	UNv1.0.ar-en.tar.gz.01
	UNv1.0.en-fr.tar.gz.02		UNv1.0.en-ru.tar.gz.02		

B. Building the Parallel Corpus

1. Downloading and Extracting the UN Corpus Files

The UN Parallel Corpus is distributed in split .tar.gz segments (e.g., .gz.00, .gz.01, .gz.02). These are standard compressed archive parts used for large datasets. A .tar file bundles many files together, and .gz compresses that bundle. Large corpora are often split into multiple numbered chunks for easier downloading.

After downloading the English and Arabic segments, the files were extracted using the terminal:

```
dohahashish@Dohas-MacBook-Air Desktop % tar -xzf UNv1.0.ar-en.tar.gz.00
x ar-en/
x ar-en/UNv1.0.pdf
x ar-en/UNv1.0.ar-en.en
x ar-en/README
x ar-en/DISCLAIMER
x ar-en/UNv1.0.ar-en.ids
x ar-en/UNv1.0.ar-en.ar: truncated gzip input: Unknown error: -1
tar: Error exit delayed from previous errors.
dohahashish@Dohas-MacBook-Air Desktop %
```

A common message appeared during extraction:

tar: Error exit delayed from previous errors

```
tar: Error exit delayed from previous errors.
dohahashish@Dohas-MacBook-Air Desktop %
```

This indicates that tar reached a corrupted or incomplete part at the end of the archive. However, the required files were already extracted successfully before the error occurred.

The extracted archive contained:

- English source file “UNv1.0.ar-en.en”
- Arabic human translation file “UNv1.0.ar-en.ar”
- Alignment ID file “UNv1.0.ar-en.ids”
- Other files like disclaimer and readme

```
dohahashish@Dohas-MacBook-Air Desktop % cd ar-en
dohahashish@Dohas-MacBook-Air ar-en % ls
DISCLAIMER      UNv1.0.ar-en.ar      UNv1.0.ar-en.ids
README          UNv1.0.ar-en.en      UNv1.0.pdf
dohahashish@Dohas-MacBook-Air ar-en %
```

2. Alignment Attempts

2.1 Alignment Using UN Alignment IDs

The UN provides an alignment ID file intended to match English and Arabic segments. A Python script was used to test this method:

```
1  en_file = "UNv1.0.ar-en.en"
2  ar_file = "UNv1.0.ar-en.ar"
3  ids_file = "UNv1.0.ar-en.ids"
4
5  with open(en_file, encoding="utf-8") as f:
6      en = [None] + [line.strip() for line in f]
7
8  with open(ar_file, encoding="utf-8") as f:
9      ar = [None] + [line.strip() for line in f]
10
11 with open("parallel_corpus.txt", "w", encoding="utf-8") as out:
12
13     with open(ids_file, encoding="utf-8") as f:
14         for i, line in enumerate(f, 1):
15             parts = line.strip().split()
16             if len(parts) < 3:
17                 continue
18
19             try:
20                 ar_idx = int(parts[1].split(":")[1])
21                 en_idx = int(parts[2].split(":")[1])
22             except:
23                 continue
24
25             if ar_idx < len(ar) and en_idx < len(en):
26                 out.write(f"{i}: {en[en_idx]}\n")
27                 out.write(f"{i}: {ar[ar_idx]}\n")
28
29 print("Done → parallel_corpus.txt")
```

Running the script produced a file named `parallel_corpus.txt`, but the alignment quality was poor. The IDs did not correspond cleanly to the English and Arabic segments, resulting in unusable output.

```
dohahashish@Dohas-MacBook-Air ar-en % head -n 20 parallel_corpus.txt
1: ON THE QUESTION OF AN INTERNATIONAL CRIMINAL JURISDICTION
1: المحتويات
2: CONTENTS
2: الدانمرك* . ٢١
3: Page
3: السويد* . ٢١
4: INTRODUCTION . 2
4: فنلندا* . ٢١
5: Australia . 3
5: المكسيك . ٢١
6: Denmark* . 17
6: النرويج . ٢٥
7: Finland* . 17
7: الولايات المتحدة الأمريكية . ٢٧
8: Iceland* . 17
8: الرد المرسل باسم بلدان الشمال الأوروبي مستنسخ بعنوان "النرويج* .
9: Mexico . 19
9: مقدمة
10: United States of America . 25
10: - اعتمدت الجمعية العامة في دورتها السابعة والأربعين، المعقودة عام ١٩٩٢، القرار ٤٧/٢٢؛ المؤرخ ٢٥ تشرين الثاني/نوفمبر ١٩٩٢ المعنون "تقرير لجنة القانون الدولي عن أعمال دورتها الرابعة والأربعين".
```

2.2 Line-Break Alignment

Because the ID-based alignment was unreliable, the English and Arabic files were aligned using line breaks. Inspection of both files showed that each line in the English file corresponds to the same line number in the Arabic file.

To verify this, both files were converted to .txt format and aligned using awk:

```
dohahashish@Dohas-MacBook-Air ar-en % paste english.txt arabic.txt | awk -F'\t' '{print NR ": " $1; print NR ": " $2}' > parallel_line_breaks.txt
```

Screenshots confirm that the first, middle, and last lines are correctly aligned.

First 40 lines:

```
dohahashish@Dohas-MacBook-Air ar-en % head -n 40 parallel_line_breaks.txt
1. ORIGINAL: ENGLISH/SPANISH
1. ORIGINAL: ENGLISH/SPANISH
2. INTERNATIONAL LAW COMMISSION
2. لجنة القانون الدولي
3. Forty-fifth session
3. الدورة الخامسة والأربعون
4. 3 May-23 July 1993
4. ٣ أيار/ مايو - ٢٣ تموز/ يوليه ١٩٩٣
5. COMMENTS OF GOVERNMENTS ON THE REPORT OF THE WORKING GROUP
5. تعليقات الحكومات على تقرير الفريق العامل المعني
6. ON THE QUESTION OF AN INTERNATIONAL CRIMINAL JURISDICTION
6. بمسألة إنشاء قضاء جنائي دولي
7. CONTENTS
7. المحتويات
8. Page
8. الصفحة
9. INTRODUCTION . 2
9. مقدمة . ٢
10. Australia . 3
10. استراليا . ٤
11. Denmark* . 17
11. الدانمرك* . ٢١
12. Finland* . 17
12. السويد* . ٢١
13. Iceland* . 17
13. فنلندا* . ٢١
14. Mexico . 19
14. المكسيك . ٢١
15. Norway . 23
15. النرويج . ٢٥
16. United States of America . 25
16. الولايات المتحدة الأمريكية . ٢٧
17. * The reply sent on behalf of the Nordic countries is reproduced under the heading "Norway".
17. "الرد المرسل باسم بلدان الشمال الأوروبي مستنسخ بعنوان "النرويج".
18. INTRODUCTION
18. مقدمة
19. 1. At its forty-fourth session, in 1992, the General Assembly adopted resolution 47/33 of 25 November 1992 entitled "Report of the International Law Commission on the work of its forty-fourth session".
19. ١ - اعتمدت الجمعية العامة في دورتها السابعة والأربعين، المعقودة عام ١٩٩٢، القرار ٤٧/٣٣ المؤرخ ٢٥ تشرين الثاني/نوفمبر ١٩٩٢ المعنون "تقرير لجنة القانون الدولي عن أعمال دورتها الرابعة والأربعين/".
20. 2. Paragraphs 4, 5 and 6 of the resolution read as follows:
20. ٢ - وفيما يلي نص الفقرات ٤ و ٥ و ٦ من القرار.
dohahashish@Dohas-MacBook-Air ar-en %
```

10 lines from the middle

```
dohahashish@Dohas-MacBook-Air ar-en % head -n 40 parallel_line_breaks.txt
1. ORIGINAL: ENGLISH/SPANISH
1. ORIGINAL: ENGLISH/SPANISH
2. INTERNATIONAL LAW COMMISSION
2. لجنة القانون الدولي
3. Forty-fifth session
3. الدورة الخامسة والأربعون
4. 3 May-23 July 1993
4. ٣ أيار/ مايو - ٢٣ تموز/يوليه ١٩٩٣
5. COMMENTS OF GOVERNMENTS ON THE REPORT OF THE WORKING GROUP
5. تعليقات الحكومات على تقرير الفريق العامل المعني
6. ON THE QUESTION OF AN INTERNATIONAL CRIMINAL JURISDICTION
6. بمسألة إنشاء قضاء جنائي دولي
7. CONTENTS
7. المحتويات
8. Page
8. الصفحة
9. INTRODUCTION . 2
9. مقدمة . ٢
10. Australia . 3
10. استراليا . ٤
11. Denmark* . 17
11. الدانمرك* . ٢١
12. Finland* . 17
12. السويد* . ٢١
13. Iceland* . 17
13. فنلندا* . ٢١
14. Mexico . 19
14. المكسيك . ٢١
15. Norway . 23
15. النرويج . ٢٥
16. United States of America . 25
16. الولايات المتحدة الأمريكية . ٢٧
17. * The reply sent on behalf of the Nordic countries is reproduced under the heading "Norway".
17. "الرد المرسل باسم بلدان الشمال الأوروبي مستنسخ بعنوان "النرويج" *
18. INTRODUCTION
18. مقدمة
19. 1. At its forty-fourth session, in 1992, the General Assembly adopted resolution 47/33 of 25 November 1992 entitled "Report of the International Law Commission on the work of its forty-fourth session".
19. ١ - اعتمدت الجمعية العامة في دورتها السابعة والأربعين، المعقودة عام ١٩٩٢، القرار ٤٧/٣٣ المؤرخ ٢٥ تشرين الثاني/نوفمبر ١٩٩٢ المعنون "تقرير لجنة القانون الدولي عن أعمال دورتها الرابعة والأربعين/٢٠. 2. Paragraphs 4, 5 and 6 of the resolution read as follows:
20. ٢ - وفيما يلي نص الفقرات ٤ و ٥ و ٦ من القرار
dohahashish@Dohas-MacBook-Air ar-en %
```

Last 10 lines

```
dohahashish@Dohas-MacBook-Air ar-en % tail -n 10 parallel_line_breaks.txt
3160510. 33. The successful conclusion of the Rome Conference represented a major development in the promotion of international law.
3160510. ٣٣ - ومضى يقول إن مؤتمر روما الذي اختتم بنجاح يمثل تطورا هاما في تعزيز القانون الدولي.
3160511. Several times in the twentieth century, the Macedonian nation had been the victim of genocide, war crimes and crimes against humanity; it was therefore natural that his Government had consistently supported the establishment of an international criminal court.
3160511. وقد وقعت الأمة المقدونية عدة مرات في القرن العشرين ضحية الإبادة الجماعية، وجرائم الحرب والجرائم ضد الإنسانية؛ ولذلك كان من الطبيعي أن تساند حكومته باستمرار إنشاء محكمة جنائية دولية.
3160512. 34. The task of implementing resolution F of the Rome Conference lay ahead. It was to be hoped that the Preparatory Commission would accomplish its tasks in a timely manner.
3160512. ٣٤ - وقال إن أمامنا مهمة تنفيذ القرار واو الصادر عن مؤتمر روما، ويؤمل أن تقوم اللجنة التحضيرية بإنجاز أعمالها في الوقت المناسب.
3160513. The most important task, however, was the promotion of the Statute, so that the Court could begin functioning at the beginning of the next century.
3160513. غير أن أكثر الأعمال أهمية هو تعزيز النظام الأساسي كي تستطيع المحكمة البدء في العمل في مطلع القرن القادم.
3160514. 35. His delegation believed that it was very important not to change the Statute of the Court as adopted.
3160514. ٣٥ - وأعرب عن اعتقاده وفده بأن من المهم جدا ألا يجرى أي تغيير في النظام الأساسي للمحكمة بصيغته المعتمدة.
dohahashish@Dohas-MacBook-Air ar-en %
```

3. Sampling the First 5,000 Lines

The full corpus contains **3,160,514 lines**, which is too large for manual MWE evaluation. Therefore, a sample of the first **5,000** aligned lines was extracted.

The last 10 lines of the sample were checked to ensure the file contains exactly 5,000 entries.

```
[dohahashish@Dohas-MacBook-Air ar-en % tail -n 10 parallel_5000.txt
4996. Base/repeater antennas
4996. هوائيات للمحطات القاعدية / التكرارية
4997. Paging systems
4997. نظم استدعاء
4998. Base antennas
4998. هوائيات قاعدية
4999. Mobile radio sets
4999. أجهزة لاسلكية متنقلة
5000. Satellite equipment
5000. معدات السواتل
```

4. Preparing the Machine Translation Column

Since the corpus is aligned by line breaks and DeepL does not support Excel uploads, **Google Translate** was used to generate the machine-translated Arabic output.

4.1 Extracting the First 5,000 English Lines

The first 5,000 English lines were extracted and saved in a TSV file, which was then imported into Excel to create:

- english_5000.xlsx

4.2 Obtaining Google Translate Output

The English lines were translated using Google Translate and saved as:

- arabic_5000_Google_Translation.xlsx

Because AntConc requires plain text input, the Google translation file was converted back into .txt format.

This produced the final three files needed for AntPConc:

1. english_5000.txt
2. arabic_5000.txt
3. google_5000.txt

These were then loaded into AntPConc to create the three-row parallel corpus used for MWE evaluation "Parallel_Corpus_5000.db".